Where is that car’s odometer right now?

Rental readings can arrive weeks late.
Valuation and defleet decisions still need mileage today.

01

Estimate today

Current mileage + uncertainty

02

Plan ahead

Future mileage for defleet

03

Review exceptions

Unusual readings + context

2,699,762 development rows • **27,630** vehicles • local, reproducible pipeline

A new status is not always a new mileage reading.

MEASUREMENT CLOCK

When the value belongs

An OnRent row may carry older handover mileage.

AVAILABILITY CLOCK

When we may use it

The value and rental metadata enter the model only after the row is recorded.

Handover / measurement — Prediction date — Row recorded

> **Retiming must never reveal future information.**
>
> Future-field mutation tests protect earlier predictions.

Evidence, with qualification: 424,008 paired vehicle-days have lower OnRent maxima; 14,660 have higher maxima. Field semantics still require confirmation.

Learn usage. Preserve time. Show uncertainty.

Last known mileage + rented-day rate × rented days + idle-day rate × idle days

1

Build the history

Separate measurement from availability.

→

2

Compare six models

Two baselines, rental-aware variants, blend, robust rate.

→

3

Deliver & review

Point estimates + ranges + explainable flags.

FIT	SELECT THROUGH	CALIBRATE THROUGH	TEST
31 Mar 2025	21 Jul 2025	11 Nov 2025	Later held-out targets

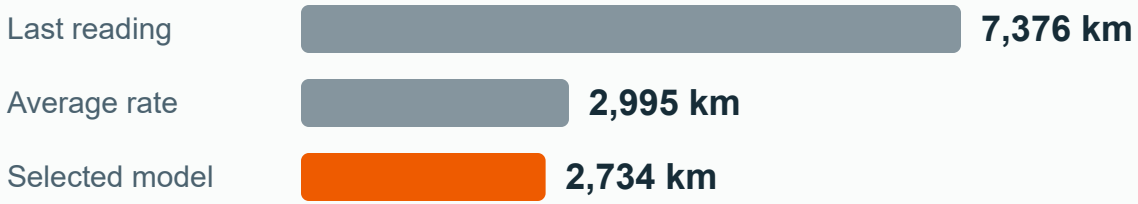
22,071 training / 5,559 held-out vehicles. Shared priors exclude held-out vehicles; their own earlier history remains available. Validation chose **recent rental-aware** nowcasts and **blended** forecasts.

Lower error than both baselines.

CURRENT MILEAGE



FUTURE MILEAGE



Selected model	Current	Future
MAE, km ↓	**415**	**2,734**
MAPE ↓	7.28%	17.31%
Range coverage*	90.7%	95.6%

26.0% / 62.9%

lower MAE vs last reading

20,732 current / 14,274 future targets.
*90% calibration target; coverage is empirical.

94.2%
Injected-error episode recall

93.9%
Exact corrupted-reading recall

84.22 / 1,000
Flagged readings; real precision pending

Targets are recorded return-source readings and may contain errors. LongTerm test groups are small. Forecast ranges are wide; better coverage alone does not establish decision value.

Sources: metrics.json · flag_benchmark.json

A daily decision aid.
A measurable next step.

What we deliver

Valuation & defleet	**Estimate + age + range**
Review operations	**Ranked signals + context**
Masked target output	**163,047 predictions**

3,745 no-history cohort fallbacks require review.
No ground truth is available for masked-target accuracy.

Open the synthetic interactive demo ↗

Ask: a shadow-mode pilot

Confirm semantics
Handover readings, planned returns and status codes.

Validate the deployed procedure
Long rentals, no-history cases and exact export settings.

Measure analyst value
Expert flag labels, review workload and range usefulness.

31 tests passed • **public example runs independently** • **no paid inference API**

Prototype evidence, not a production guarantee. Data publication permission must be resolved before sharing organizer-derived material.